THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force can NOT: 1

(a) change shape **(b)** change direction
(c) change the mass of an object **(d)** start motion

2. Which of the following is a contact force? 1

(a) magnetic force **(b)** gravitational force
(c) electrostatic force **(d)** friction

3. Friction is greater on a: 1

(a) polished glass **(b)** smooth surface **(c)** rough surface **(d)** ice

4. Unlike charges: 1

(a) melt **(b)** cancel mass **(c)** repel **(d)** attract

5. Gravitational force is always: 1

(a) repulsive **(b)** contact-based **(c)** attractive **(d)** zero

6. A force of 900 N acts perpendicularly on an area of 3 m². The pressure is: 1

(a) 600 N/m² **(b)** 300 N/m² **(c)** 150 N/m² **(d)** 2700 N/m²

7. A knife cuts better with a sharp edge because: 1

(a) it is magnetic **(b)** it is heavier
(c) larger area gives more pressure **(d)** smaller area gives more pressure

8. The base of a dam is broader than its top because: 1

(a) it looks better **(b)** water is lighter below
(c) pressure is largest near the bottom **(d)** pressure is zero below

9. A rubber sucker sticks to a wall because: 1

(a) outside air pressure is more than inside **(b)** glue is present
(c) friction only **(d)** magnetism

10. Wind speed is greater when the pressure difference is: 1

(a) zero **(b)** smaller **(c)** negative **(d)** greater

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is upthrust? In which direction does it act? 2

12. State Archimedes' principle. 2

13. What causes lightning? 2

14. What causes thunder? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why does a ship float while an iron nail sinks? 3

16. What is the practical unit of atmospheric pressure? Convert 1013 hPa to Pa. 3

17. A body is hung from a spring balance and read as 8 N in air and 6 N in water. Find the upthrust. 3

OR

Is gravitational force a contact force? Justify. 3

18. Explain why a flat roof can be lifted in a strong wind. 3

OR

Why do cyclones cause long-term harm to agriculture? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Describe the experiment to measure upthrust using a spring balance and a stone. 5

OR

Why do we say forces are interactions? Explain with pushing a table and two more examples. 5

20. Explain local storms of India: Kalboishakhi, Bordoila and mango showers. State their usefulness. 5

OR

Calculate pressure for 2 situations: a 40 N brick on its 0.1 m² face and on its 0.05 m² face. Explain the difference. 5

END OF QUESTION PAPER — Total: 40 Marks

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